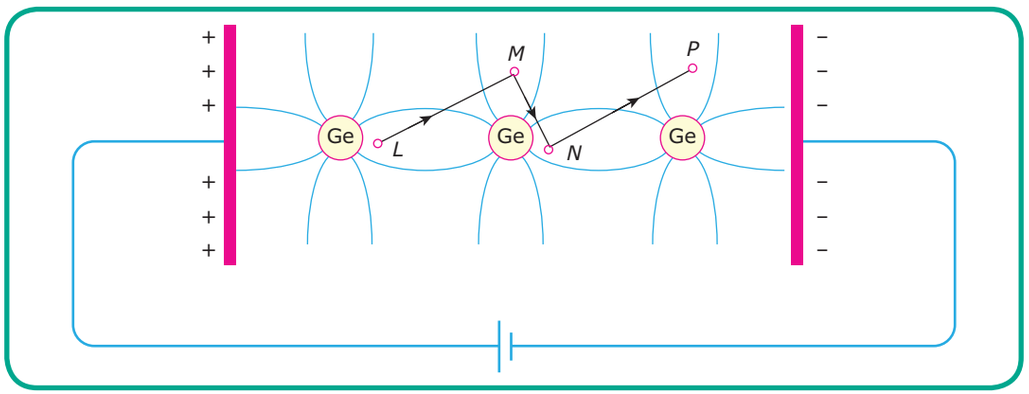


Figure 3.19 Electron and hole current formation in Ge

Though the hole current is happening due to the movement of electrons from one covalent bond to another bond, it is quite understandable why to call it as hole current. The basic reason for current flow is the presence of holes in the covalent bonds. Therefore, it is more appropriate to consider the current as the movement of holes.

3.6.7 Energy Band description of Hole Current

The hole current can be beautifully explained in terms of energy bands. Suppose due to thermal energy, an electron leaves the valence band and enter into the conduction band as shown in Figure 3.20.

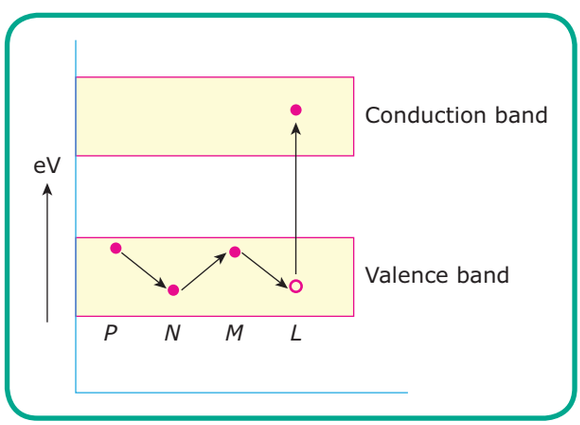


Figure 3.20 Hole movement across the band

This leaves a vacancy at L. Now, the valence electron at M comes to fill the hole at L. The result is that hole disappears at L and appears at M. Next, the valence electron at N moves to hole at M, consequently a hole is created at N. It is clear that valence electrons move along the path PNML whereas holes move in opposite direction i.e., along the path LMNP as shown in Figure 3.20.

3.7 INTRINSIC SEMICONDUCTOR

A semiconductor in an extremely pure form is known as an intrinsic semiconductor. In an intrinsic semiconductor, even at room temperature, hole-electron pairs are created. When electric field is applied across the semiconductor, the current conduction takes place by two processes such as (i) by free electrons and (ii) by holes as shown in the Figure 3.21. The free electrons are produced due to the breaking up of some covalent bonds by thermal energy.

At the same time, holes are created in the covalent bonds, under the influence of electric field. Thus, the conduction in semiconductors is by both electrons and

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